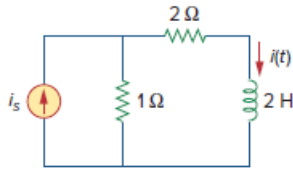


Problem

Find $i(t)$ in the circuit of Fig. 17.68 given that

$$i_s(t) = 3.5 + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos 3nt \text{ A}$$

Figure 17.68:



Step-by-step solution

Step 1 of 3

Refer to Figure 17.68 in the textbook.

Consider the expression for source current, $i_s(t) = 3.5 + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos 3nt \text{ A}$.

The value of ω_n is, $\omega_n = 3n$.

Apply current division rule.

$$\mathbf{I} = \frac{1}{1 + 2 + j\omega 2} i_s$$

Substitute $3n$ for ω_n in the equation.

$$\mathbf{I} = \frac{1}{3 + j2(3n)} i_s$$

$$= \frac{1}{3 + j6n} i_s$$

$$= \frac{1}{3(1 + j2n)} i_s$$

$$\mathbf{I} = \frac{1}{3\sqrt{1 + 4n^2} \angle \tan^{-1}(2n)} i_s \dots\dots (1)$$

Step 2 of 3

For dc component, $\omega_n = 0$ or $n = 0$.

$$I_{dc} = \frac{1}{3\sqrt{1 + 4(0)^2} \angle 0^\circ} \times 3.5$$

$$= \frac{1}{3}(3.5)$$

$$= \frac{7}{6} \text{ A}$$

Step 3 of 3

For the n th harmonic, determine the expression for $i_s(t)$.

$$i_s(t) = 3.5 + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos 3nt$$

$$i_s = \frac{4}{n^2} \angle 0^\circ$$

Substitute the expression of i_s in equation (1).

$$\begin{aligned} \mathbf{I} &= \frac{1}{3\sqrt{1+4n^2} \angle \tan^{-1}(2n)} \left(\frac{4}{n^2} \angle 0^\circ \right) \\ &= \frac{4 \angle -\tan^{-1}(2n)}{3n^2 \sqrt{1+4n^2}} \end{aligned}$$

The time domain expression is,

$$\begin{aligned} i(t) &= \mathbf{I}_{dc} + \sum_{n=1}^{\infty} \mathbf{I} \\ &= \frac{7}{6} + \sum_{n=1}^{\infty} \frac{4}{3n^2 \sqrt{1+4n^2}} \cos(3nt - \tan^{-1}(2n)) \text{ A} \end{aligned}$$

Therefore, the time domain expression of the current is

$$\boxed{i(t) = \frac{7}{6} + \sum_{n=1}^{\infty} \frac{4}{3n^2 \sqrt{1+4n^2}} \cos(3nt - \tan^{-1}(2n)) \text{ A} .}$$